

Matthew Oj

+447548238672 | edafeoj@gmail.com | [linkedin.com/in/matthew-oj](https://www.linkedin.com/in/matthew-oj) | github.com/Matthew-Oj | matthew-oj.github.io/

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript/TypeScript, PHP, HTML/CSS

Frameworks & Tools: FastAPI, Next.js, React, Astro, Streamlit, Laravel, Docker, GitHub Actions

Cloud & Infrastructure: AWS (Lambda, EC2, S3), GCP (Cloud Run, Vertex AI), Vercel, Netlify, CI/CD

Data & Databases: PostgreSQL, SQLite, Snowflake, BigQuery, dbt, Apache Airflow, Snowpipe, MongoDB, DuckDB

ML & AI: XGBoost, SHAP, Snowflake Model Registry, Jupyter Notebooks, Snowflake Notebooks, Google Colab, Vertex AI Generative Models

WORK EXPERIENCES

Senior Data Engineer (Software Engineering & Backend Infrastructure)

Apr 2026 – present

VAFO Digital

London, UK

- Designed and led migration of backend orchestration from GitHub Actions to a self-hosted Apache Airflow service on AWS EC2 (Docker Compose), replacing fragmented CI-based scheduling with a unified production data platform supporting 100+ pipelines/DAGs
- Led the end-to-end ML pipeline builds with Data Science (dbt, XGBoost/SHAP, Snowflake Model Registry), orchestrated via Airflow on EC2, powering marketing spend decisions across 2 brands and 60+ campaigns.
- Architected a distributed reporting service (Python, FastAPI, AWS Lambda, Mangum) serving automated scheduled delivery across 10 client accounts, eliminating manual data requests and cutting ad hoc engineering overhead.
- Built a shared Python library Built a shared Python library deployed as an AWS Lambda Layer, standardising error handling and logging across 30+ production workflows and 4 integration targets (Klaviyo, DynamoDB, SFTP, S3), reducing duplicate code.

Data Engineer

Apr 2025 – Apr 2026

VAFO Digital

London, UK

- Architected a monorepo data platform (dbt, Snowflake, GitHub Actions CI/CD) supporting 6+ isolated client environments with automated testing.
- Built a high-performance internal analytics platform (Python, Streamlit, Snowflake) processing 200M+ events, using warehouse clustering and query caching to deliver real-time insights to non-technical stakeholders.
- Developed Python AWS Lambda services syncing transformed Snowflake models to external systems (HubSpot, Klaviyo), supporting 250K+ users and 1K+ events/day.
- Engineered a PHP/Laravel ingestion system (AWS S3, Snowpipe) processing 10M+ records/day from REST APIs, enabling near real-time analytics.

Software Engineer (Freelance) maisonprive.london

Apr 2026 – June 2026

Maison Privé Concierge Limited

Remote

- Built and deployed a luxury concierge marketing site (Astro, TypeScript) on Vercel, with SSR API routes handling enquiries across 9 services.
- Engineered a dynamic enquiry form with per-service field schemas across 3 categories, validation, rate limiting, and spam filtering, blocking 95%+ of bot submissions.
- Built a dual-sink pipeline routing enquiry to Mailgun and Airtable in real time, plus a cached editorial feed cutting load times by 80%.

Data Engineer

Nov 2023 – Apr 2025

EssenceMediacom

London, UK

- Built a REST API-driven alert system (Python, Cloud Run) detecting Adverity DataStream failures in near real-time, enabling fast incident response and reducing missed pipeline failures.
- Designed and built a GCP-hosted web application (Python, Cloud Run) automating table schema generation for data pipelines, cutting manual pipeline build time by 30% across the analytics team.
- Optimised CI/CD deployment pipeline for Apache Airflow (GitHub Actions, Docker) DAGs, reducing deployment time by 40% and enabling faster iteration on 40+ production pipelines.

IT Support Analyst

Jan 2021 – Oct 2023

University of Greenwich

London, UK

- Provided first-line IT support across 3 campuses, managing 100+ assets and resolving 20–30 daily tickets while building internal HTML/CSS tooling to streamline helpdesk workflows.

PROJECTS

Personal Finance Platform | FastAPI, dbt, PostgreSQL, Next.js, Docker | [GitHub](#)

April 2026

- Designed a production-style data stack ingesting raw financial data through dbt models into aggregated PostgreSQL marts, served via a JWT-secured FastAPI and visualised in a Next.js dashboard, all containerised with Docker Compose.
- Wrote a pytest suite covering auth, token validation, all endpoints, and pagination against a fully mocked database - 13/13 tests passing.

WPP & Google Cloud Hackathon (3rd Place) | Python, Cloud Run, Vertex AI, Vertex AI Generative Models

Sep 2024

- Designed and deployed a Cloud Run service integrating multiple APIs with Vertex AI to automatically generate real-time advertising creatives at scale - placing 3rd out of all competing teams.

EDUCATION

University of Greenwich

Sep 2020 – May 2023

Bachelor of Engineering, Civil Engineering, 2.1 Upper Second-Class Honours

London, UK

Dissertation (First Class): Assessing the vulnerability of catchments during flash flood events in South England using Machine Learning methods